

Nestbloq Rental Management System

1. System Overview & Target Roles

The **Nestbloq Rental Management System** is a unified, end-to-end cloud platform designed for landlords, property managers, residents, and service contractors. It automates the complete rental lifecycle—from property inventory management and automated tenant screening to electronic lease execution, recurring ledger billing, online payment processing, and maintenance dispatching.

Target Role	Key Operational Capabilities
Landlord / Property Manager	Configures properties & units, invites applicants, reviews screening reports, drafts/signs leases, manages ledgers, and dispatches vendor repair requests.
Tenant / Resident	Submits rental applications, consents to background checks, e-signs lease agreements, views invoices, pays rent online (ACH/Card), and opens maintenance tickets.
Maintenance Vendor	Registered by landlords with verified license & insurance credentials. Accepts work order dispatches and submits repair cost estimates.
Superadmin	Oversees platform-wide operations, manages system parameters, and reviews audit logs and financial analytical reporting.

2. Core Modules & Detailed Use Cases

Module A: Property & Unit Inventory Setup

UC 1.1: Multi-Unit Property Setup Wizard

- **Actor:** Landlord / Superadmin
- **Workflow:** Landlord inputs Property Name, Address, City, State, and ZIP. Address validations enforce US state formatting and zip code standards.
- **Nested Execution:** Landlord adds multiple units (Unit Number & Base Rent) simultaneously. System initializes unit statuses as *VACANT* and *ACTIVE*.

Module B: Tenant Screening & Automated Background Checks

UC 2.1: Tenant Screening Invitation

- **Actor:** Landlord
- **Workflow:** Landlord selects a vacant unit and issues an email invitation to the applicant. System generates a secure registration URL.

UC 2.2: Application Submission & Automated Screening Engine

- **Actor:** Tenant Applicant
- **Workflow:** Applicant fills employment status, monthly income, pet details, and personal references. Upon submission, the background check engine automatically evaluates:
 1. **FICO Credit Score:** Algorithmic calculation based on income-to-rent ratio and employment stability.
 2. **Criminal History:** Automated query against federal and state criminal registries.

- 3. **Eviction History:** Automated search against county civil court eviction records.

UC 2.3: Landlord Application Decisioning

- **Actor:** Landlord
- **Workflow:** Landlord reviews background check reports in the Screening Hub and sets application status to *APPROVED* or *REJECTED*.

Background Screening Simulation Modes:

The Nestbloq backend engine supports instant simulation parameters for evaluation:

- **CLEAN:** Simulates background checks with high credit score & clear history.
 - **CRIMINAL:** Simulates flagged criminal registry records (e.g., Larceny, Burglary charges).
 - **EVICTED:** Simulates civil court judgment records for non-payment of rent.
- This allows evaluation of how high-risk applicants are automatically screened out.*

Module C: Digital Lease Agreements & E-Signatures

UC 3.1: Lease Creation & Custom Financial Terms

- **Actor:** Landlord
- **Workflow:** Landlord selects an approved application and configures start/end dates, base rent, security deposit, and ancillary recurring fees (Utilities, Parking, Pet Fee).
- **Penalty Terms:** Landlord specifies Grace Period days and Late Fee rules (Flat dollar amount or Percentage rate).

UC 3.2: Multi-Party Electronic Signing

- **Actor:** Landlord, Tenant, Co-Landlord
- **Workflow:** The system provides full contract review. Landlord, Tenant, and optional Co-Landlord digitally sign. Upon final signature, status becomes *ACTIVE* and unit becomes *RENTED*.

Module D: Automated Rent Ledger & Payment Gateway

UC 4.1: Automated Monthly Statement Engine

- **Workflow:** On the 1st of each month, the engine iterates over active leases and generates an itemized invoice:
Total Invoice = Rent Amount + Utilities Fee + Parking Fee + Pet Fee

UC 4.2: Automated Late Fee Enforcement Engine

- **Workflow:** If an invoice remains unpaid past Due Date + Grace Period, the engine automatically converts status to *OVERDUE* and applies the configured late fee (Flat or Percentage). Landlords retain manual override/revert capabilities.

UC 4.3: Resident Online Payment Processing

- **Actor:** Tenant
- **Workflow:** Tenant views itemized ledger statement and submits payment via **ACH** or **Credit Card**. System updates invoice to *PAID* and records an audit transaction reference (txn_XXXXXXX).

Module E: Maintenance Desk & Vendor Dispatch

UC 5.1: Tenant Repair Service Request

- **Actor:** Tenant

- **Workflow:** Tenant logs a service request with Title, Description, and Priority level (*LOW, NORMAL, HIGH, URGENT*).

UC 5.2: Contractor Credential & Compliance Verification

- **Actor:** Landlord / Vendor
- **Workflow:** Landlord registers contractors with license number, insurance policy, and expiration dates. System validates compliance to protect against liability.

UC 5.3: Work Order Dispatch & Invoice Resolution

- **Actor:** Landlord / Tenant
- **Workflow:** Landlord assigns ticket to a verified vendor, inputs repair cost estimates, and closes ticket upon completion. System creates a maintenance payment charge.